

P1: Database Design and Data Modelling

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Introduction and Overview

The real-world system that we have decided to model is a museum. We will abstract this by focusing on **Artifacts** as our central entities, since they represent the core assets of the museum and the primary objects of interest in its collection. Artifacts are made by **Artists**, and will have subtypes such as **Paintings**, **Photographs** and **Sculptures**, allowing us to capture the different medium-specific attributes. Artifacts can also be categorized into different **Genres**.

In addition to collection management, the museum is a reservation-based system. Our database will have to manage incoming **Visitors** that will reserve **Tour Dates** for a specific **Tour**, which will have a specific time and date. These tours are organized and guided by **Museum Personnel**, and consist of ordered paths between the different **Exhibitions** available at the museum.

To interact with the museum database, our application must focus on serving the needs of both visitors and museum personnel. On one hand, the system must facilitate the management of collections of artifacts, which involves being able to group them and have a complex categorization system that allows interesting filtering. On the other hand, it must allow museum personnel to create new tours, and manage the entire user and reservation system.

Data requirements

The real-world entities that will be central to our museum are artifacts.

Artifact: The main unit of our system, an artifact is any artwork that belongs to the museum. It will be divided into subtypes to allow special relationships with the subentities. Artifacts will frequently be a part of an exhibition, but could potentially be in storage so this is not forced. The main fields will be an artifact id, the name, the creation date (year), and description.

Painting: a subtype of artifact, this will have additional attributes that only belong to paintings such as medium on which the painting is created (canvas, paper, wood, etc.) and the type of paint used (acrylic, water color, oil, graphite pencil, etc.).

Photographs: a subtype of artifact, we will consider a photograph to be an artifact that has been digitally produced in 2 dimensions. Attributes include the film type (black and white, colour, etc.).

Sculptures: a subtype of artifact, a sculpture is a three-dimensional artifact. This includes statues, jars, crafts, etc. Attributes include material (granite, marble, wood, silicon, etc), and dimensions.

Artist: An artist is someone that has created an artifact. We will keep track of their name, date of birth, and an artist id.

Genres: Genres represent the artistic categories used to classify artifacts and exhibitions. We keep track of its genre name, such as impressionism, contemporary, renaissance, etc.

The museum will be organized into exhibitions. Guides will guide visitors through exhibitions made up of artifacts.

Exhibition: an assembled group of artifacts, curated by our experts that often have a common theme or motive. The attributes include exhibition id, name, description, start date, and end date.

Tour: a tour is a pre-defined path within the museum that visits certain exhibitions. The tours will have an assigned guide, a description, a cost, and tour id.

TourDates: TourDate represents a scheduled instance of a tour, with the attribute date, start_time, and end_time indicating the exact date and time the tour occurs. It is modelled as a weak entity, as it cannot exist independently and depends on the Tour it is associated with.

Museum_Personnel: Museum_Personnel is a person who works in the museum. They are involved in organizing and guiding visitors through tours. The attributes include employee id, and name.

On the business side of the museum we will work with a reservation system.

Visitor: a person that comes to the museum to be a part of a tour. They will have an account with the museum, and we must keep track of their name, date of birth, email. Since emails are naturally unique, this will be the primary key. They will be able to reserve tour dates allowing them to participate in a tour.

There will be many real-world relationships that we have to model:

Artifact ISA Hierarchy: To model different types of artifacts, we use an ISA hierarchy with the subclasses Painting, Photographs and Sculptures. This is non-covering, since those rare, uncommon artifacts that do not fit in any of the subcategories will be a part of the superclass Artifact.

Creates: For every artifact in the museum, there is an artist behind it. Each artifact must be created by at least one artist, but collaborations are allowed. This relationship is many to many with a participation constraint, meaning an artist can create many artifacts, but each artifact is created by one or more artists.

Displays: Artifacts are displayed in exhibitions for visitors to see. We track the room of the exhibition in which the artifacts are displayed. This one to many relationship is modelled with a key constraint, such that an exhibition can display multiple artifacts, but a specific artifact can only be shown in at most one exhibition at a time or not be displayed at all, in which case it is placed in storage.

Made_Of: Tours are made up of exhibitions that are presented to the visitors. Each tour must be made up of at least one exhibition, and each exhibition can be part of many different tours.

Leads: The tours being offered to visitors are led by museum personnel, and they will act as guides to the visitors through the exhibitions. This is presented in a one to many relationships with both key and participation constraints, meaning each tour must be led by exactly one

museum employee, and every museum personnel can guide multiple tours if there are no time conflicts.

Organizes: Tours are organized by museum personnel. A museum staff member can organize many tours, and each tour must be organized by at least one staff member.

Of: The Of relationship connects TourDates to the Tour they belong to. Each tour date/start time belongs to exactly one tour, and each tour can have multiple scheduled tour dates.

Reserves: When a visitor decides to attend a tour, they can make a reservation for a tour that is scheduled on a particular tour date. Each tour date can be reserved by multiple visitors, and each visitor can make many reservations. We track the date the reservation for a tour was made.

Categorized_as: Exhibitions and Artifacts are categorized as certain genres. This is a many to many relationship with a participation constraint, such that an exhibition and an artifact must belong to at least one genre.

Functional requirements

There will be two main users for our application: Visitors and Museum Personnel.

Landing page: Accessible without an account by any user

1. Users should be able to access the main Artifact database and make filters for time period, type of artifact, artists, etc. Should also be able to search for specific artifacts with a search bar.
2. Users should be able to look at what tours are currently available and basic information on them (description, time, date, etc).

From the visitors side:

1. Visitors should be able to create an account. This would register them as a new user in the Visitor database. This will work by inputting their information and submitting it after input validation.
2. Visitors should be able to make a reservation for a Tour Date. This will work by selecting an available timeslot (preventing overbooking with special constraints), selecting the tour they wish to attend, and finally generating a reservation ticket.

Form the Museum Personnel side:

1. Museum Personnel should be able to register in the same way as visitors.
2. Museum Personnel should be able to create new exhibitions by inputting the information for them (name, description, artifact selection etc)

These functional requirements are supported by the Visitor, Reservation, TourDates, Exhibitions, and Artifact entities and their associated relationships.

Unique aspects, constraints, restrictions and special requirements

- The start date of an Exhibition must be before its end date, and artifacts may only be displayed in exhibitions during this active period.
- Tour Dates represent scheduled tours and must not overlap in time if the same museum personnel is guiding it. This cannot be enforced in ER diagram and requires additional application logic
- Visitors can only book Tour Dates that happen after the reservation booking date.
- Each tour date may be limited to a maximum capacity. This would also be enforced at the application level.
- Artifacts can only be displayed in one exhibition at a time, or in storage. This reflects how a real museum works.
- Tours can only include exhibitions that are active at the time of the tour date.
- Museum personnel can both organize and guide tours.

Relational translation

In the form of Relation_name(attr1, attr2, attr3, ...)

Artifacts(artifact_id, name, description, date_of_creation)

Paintings(artifact_id, medium, paint_type) foreign key artifact_id references Artifacts

Photographs(artifact_id, film_type) foreign key artifact_id references Artifacts

Sculpture(artifact_id, material, dimensions) foreign key artifact_id references Artifacts

Exhibitions(exhibition_id, name, description, start_date, end_date)

Display(artifact_id, exhibition_id, room) foreign key artifact_id references Artifacts and foreign key exhibition_id references Exhibitions

Artists(artist_id, name, birth)

Creates(artifact_id, artist_id) foreign key artifact_id references Artifacts and foreign key artist_id references Artists

Genres(genre_name)

Exhibition_Genre(exhibition_id, genre_name) foreign key exhibition_id references Exhibitions and foreign key genre_name references Genres

Artifact_Genre(artifact_id, genre_name) foreign key artifact_id references Artifacts and foreign key genre_name references Genres

Tours(tour_id, description, cost, employee_id) foreign key employee_id references

Museum_Personnel

Tour_Exhibitions(exhibition_id, tour_id)

Museum_Personnel(employee_id, name)

Organizer(employee_id, tour_id) foreign key employee_id references Museum_Personnel and foreign key tour_id references Tours

TourDates(date, start_time, tour_id, end_time) foreign key tour_id references Tours

Visitor(email, name, address, birth)

Reservation(email, booking_date, tour_id, date, start_time) foreign key email references Visitor, foreign keys (date, start_time) references TourDates

Creativity and Design Highlights

Our design might stand out from other projects due to the unique idea of modelling an artifact-centric museum. This allows us to really have a complex, interesting core and a theme that affects all our design decisions. We are using both IS-A relationships and weak entities, and have a dual role in the visitors and museum personnel. We have described in detail what data is required in order to represent these real world objects, with business constraints such as artifacts being able to be exhibited or in storage, and how a real museum is organized and operated.